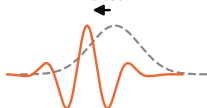


A: charge algebra $\mathcal{C}_\omega[G]$ on the bonds;
central extension $[\omega] = \text{SPT index}$

Ward identity
+ contact term

superselection
bookkeeping

2 sites



C: soft theorem

$$\delta = 2k_s + \frac{v_h}{\omega_h} k_s^2 + \dots$$

DC flux limit

$\delta x = -2T$



B: quantized memory

$$T = 1 - [1 + (4\omega v/J_\perp^2)^2]^{-1}$$